Answer all the questions.

1- **Write the main goal of distributed system** with describing the meaning of each one?

2- Define the following:

a- **Resource Sharing** with writing types of resources
b- **Transparency** and state what it hid
c- **Openness** and its Benefits
d- **Scalability** and its Techniques
e- **Reliability** and the Techniques for Dealing with Failures with detecting the failure type
f- **Performance** and its Metrics

3- State the **types of transparency** and show what each type hides (with giving an example for each one).

4- Describe the reason of Trade-offs Between **(Scalability vs. Security)** and (**Openness vs. Performance**).